

Fundamental Grid Definition (Required)

PURPOSE:

*GRID defines the type of fundamental (main) grid and marks the beginning of the reservoir description.

FORMAT:

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*GRID (*CART)   $n_i$   $n_j$   $n_k$   
          (*VARI)   $n_i$   $n_j$   $n_k$   
          (*CORNER)  $n_i$   $n_j$   $n_k$   
          (*RADIAL)  $n_i$   $n_j$   $n_k$  *RW  $r_w$ 
```

DEFINITIONS:

*CART

Specify a Cartesian (rectangular) grid type with layers of uniform thickness. If there is no tilt, the layers have uniform depth.

*VARI

Specify a rectangular grid type that allows layers of variable thickness located at varying depths.

*CORNER

Specify a corner-point grid type, which allows flexible locations for block corners (nodes) to define non-standard shapes and distribution of blocks.

*RADIAL

Specify a cylindrical (radial-angular) grid type.

n_i n_j n_k

Number of grid blocks in the I, J and K directions, respectively.

*RW r_w

The radius of the wellbore (m | ft) located inside the innermost radial blocks, for the ***RADIAL** grid type. Since $r_w = 0$ triggers a non-zero default, use a small non-zero value to model a no-wellbore case.

DEFAULTS:

Required keyword. No defaults.

If ***RW** is absent or $r_w = 0$, $r_w = 0.0762$ m (0.25 ft) is assumed.

CONDITIONS:

This keyword should be the first keyword in the Reservoir Description section.

EXPLANATION:

The keyword ***GRID** indicates the grid type and the number of grid blocks within the fundamental (main) grid system. There is no default grid type, and one of the ***GRID** sub-keywords must be specified.

The ***GRID** keyword defines a grid consisting of $n_i \times n_j \times n_k$ blocks. The blocks are rectangular in shape for Cartesian and variable depth/variable thickness grids. They should be close to rectangular for corner point grids. Radial-angular cylindrical grids consist of stacked circular rings, where the rings may have angular subdivisions.

Under ***VARI**, a full flow connection is maintained between adjacent blocks in the same layer (K index) even if they lie at different depths; this technique represents varying depths in the reservoir.

A wellbore volume of radius r_w is removed from the innermost ring in all layers of ***RADIAL** grids. This gives the innermost ring a circular inner boundary that connects to the wellbore. The wellbore volume is not included in the volume of any of the active blocks.

I, J, and K indices are used to identify the blocks, where I runs in the range of 1 to n_i , J runs in the range of 1 to n_j and K runs in the range of 1 to n_k . The notation (I,J,K) is used to address a single block.

For Cartesian and variable depth/variable thickness grids, I corresponds to the "x" direction, J to the "y" direction, and K to the "z" direction, where "x", "y" and "z" refer to a standard (right handed) coordinate system in the reservoir.

Corner point cells have all their corners specified in terms of an "x-y-z" triple defined by the incoming data. However, the I-J-K numbering system is still important because the default search methods for determining connections assume that incrementing the I or J index for each cell, and searching the resulting stack of potential neighbours by sweeping through K, should find all contacting cells. Thus, I-J planes should roughly correspond to geological layers indexed by K, even if the I axis and the "x" axis, or the J axis and the "y" axis, are not aligned.

For radial-angular cylindrical grids, I corresponds to the "r" (radial) direction, J to the " θ " (angular) direction, and K to the "z" (axial) direction.

For grid types other than *CORNER, the orientation of K direction (upward or downward) relative to gravity is determined by keyword *KDIR.

Summary of Grid Type Comparison

Here is a summary of the comparison of the different grid types.

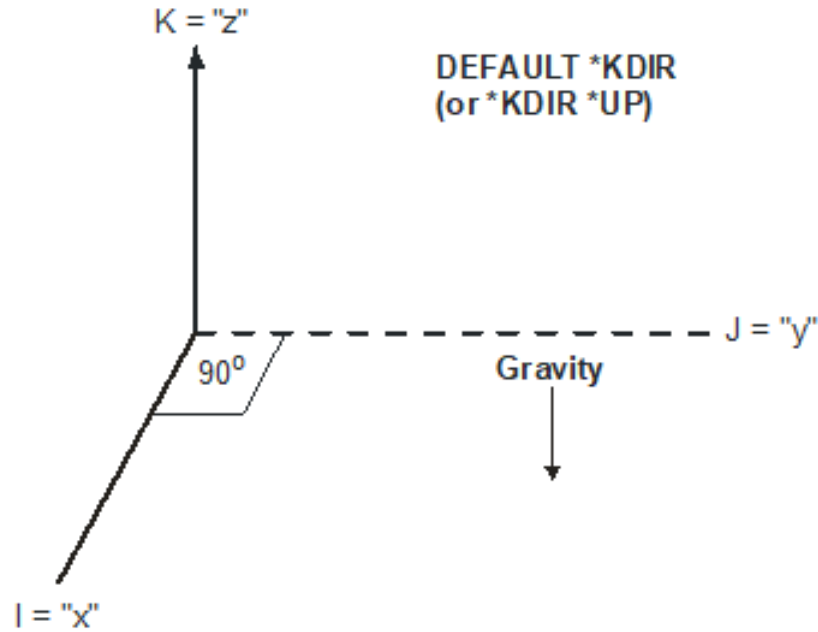
Grid Type	*CART	*VARI	*RADIAL	*CORNER
I index direction:	X	X	R	X
J index direction:	Y	Y	θ	Y
K index direction:	Z	Z	Z	Z
K index ordering:	*KDIR	*KDIR	*KDIR	Downward
*DIP allowed?	Yes	No	Yes	No
I-dir cell size	*DI	*DI	*DI	See Corner Point
J-dir cell size	*DJ	*DJ	*DJ	See Corner Point
K-dir cell size	*DK	*DK	*DK	See Corner Point
*DEPTH allowed?	Yes, rec.	Yes	Yes, rec.	No
*DTOP allowed?	Yes	Yes, rec.	Yes	No
*DEPTH-TOP allowed?	Yes	Yes, rec.	Yes	No
*PAYDEPTH allowed?	Yes	Yes, rec.	Yes	Yes
rec. - recommended				

Rectangular

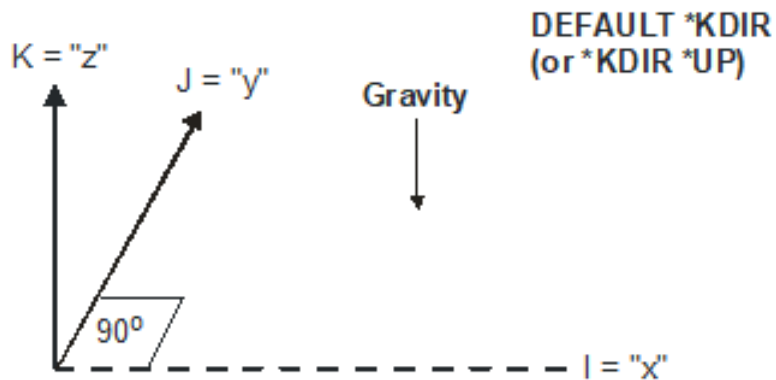
If a Cartesian or variable depth/variable thickness grid is examined with the I index increasing towards the viewer, J should increase from left to right and K should increase upwards, provided that *KDIR has not been used (or *KDIR *UP is set). If a Cartesian or variable

depth/variable thickness grid with ***KDIR *DOWN**, or a corner point grid, is examined with the J index increasing towards the viewer, I should increase from left to right and K should increase downwards.

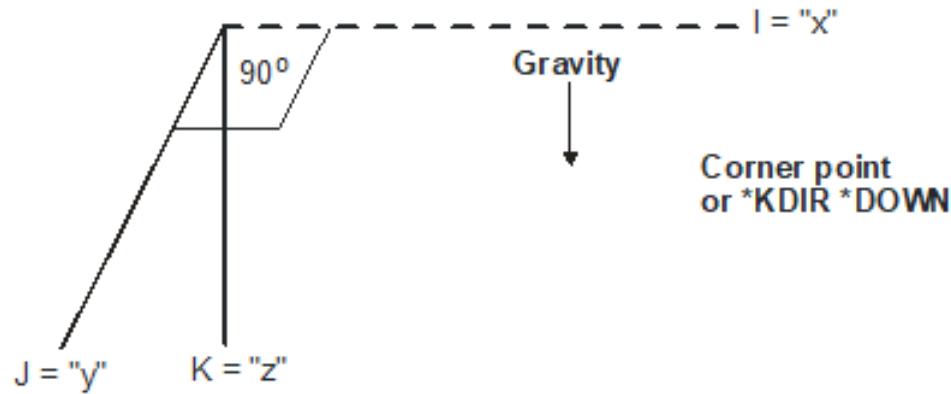
The coordinate system for a Cartesian or variable depth/variable thickness grid with the no ***KDIR** specified (default), or ***KDIR *UP** specified, appears as follows:



or viewing the above from a different vantage point:



The coordinate system for a Cartesian or variable depth/variable thickness grid with ***KDIR *DOWN** specified, or a corner point grid, appears as follows:

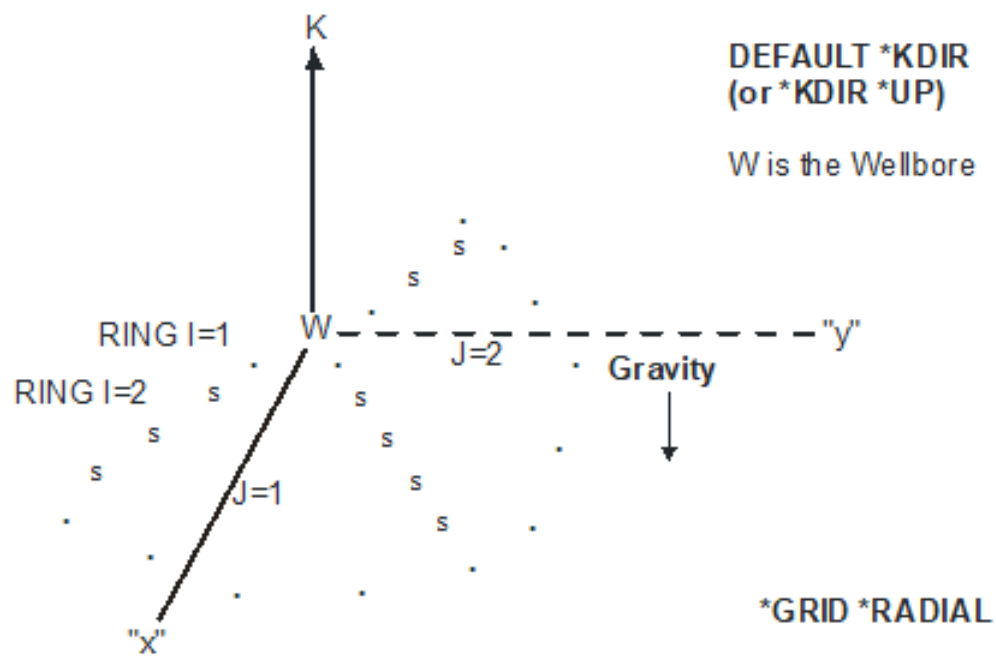


Radial

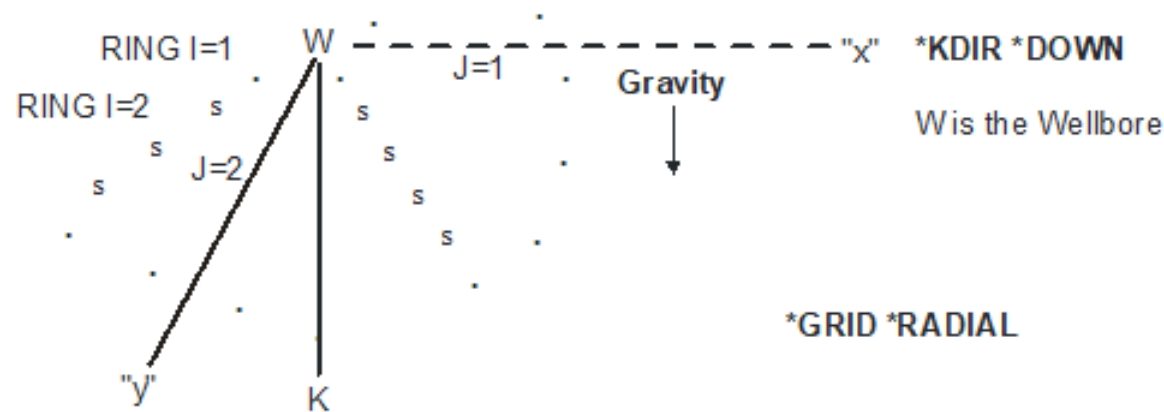
***RADIAL** radial-angular cylindrical grids have rings indexed by I , counting from the innermost outwards. Each ring may be split into n_j sectors indexed by J , counting counterclockwise around the K axis with the first sector lying over the I axis. Layers are indexed by K , starting at the reservoir bottom and counting upwards if no ***KDIR** keyword appears (or if ***KDIR *UP** is set), and counting downwards if ***KDIR *DOWN** is set.

Note that the first ring of ***RADIAL** grids is not subdivided even if n_j exceeds 1, and will exclude a volume of radius r_w ; the latter volume is not assigned an I index value. Thus, the inner radius of the first ($I=1$) grid block is given by r_w and its outer radius is given by the sum of r_w and the length of the first block in the I direction.

The coordinates for a radial-angular cylindrical grid with the no ***KDIR** specified (default), or ***KDIR *UP** specified, appears as follows:



and when ***KDIR *DOWN** is set:



For the ordering of grid blocks, See Appendix B, [Figure 1](#) for examples of a Cartesian grid system; [Figure 2](#) for a cylindrical system and [Figure 5a](#) and [Figure 5b](#) for a variable thickness/variable depth grid system.

A block's pressure node is where the pressure is evaluated, and for a rectangular block this is at the block's geometric centre. In contrast, pressure nodes for ***RADIAL** blocks are placed in accordance with an assumption made about the discretization of the accumulation terms

in the simulator. A hypothetical steady-state flow of a radially-directed incompressible fluid through the block gives a pressure distribution that varies with radius. The pressure node is located at the radius where this distribution's pressure is equal to average of the pressure distribution over the block's radial extent. Details can be found in [Pedrosa and Aziz, 1986](#).

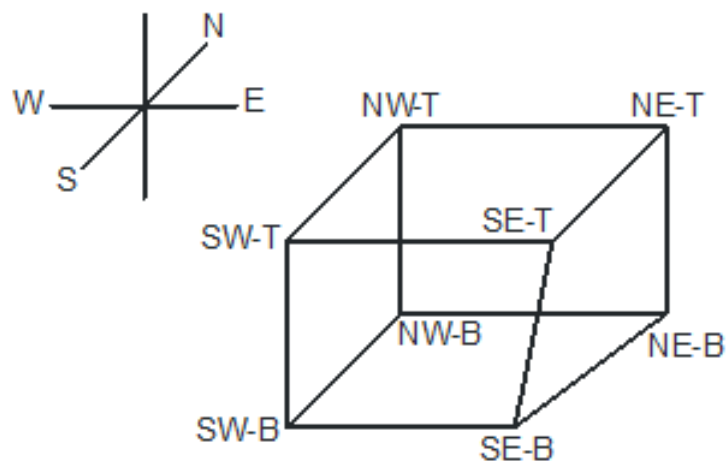
Corner Point

Corner point grids are made up of blocks which are defined by their eight corner points. Each corner point is described by giving its three coordinates, an "x"-, "y"- and "z"-coordinate, locating it in the subsurface. The "x"- and "y"- coordinates have usually been projected against a horizontal reference surface. The "z"- coordinate is usually the depth of the corner point measured downwards from that surface. Both positive or negative depths are valid, depending on the location of the reference surface ($z=0$) with respect to the reservoir, although positive values are most common.

Generally 24 numerical values are required to determine a corner point block (3 directions for 8 corners). However, some keywords place assumptions on the corner point data so that less values per block are required. The following table shows the different methods by which corner points can be assigned. Here n_{nodes} is the number of unique (non-colocated) corners, which for a *CART grid type corresponds to $(n_i+1) (n_j+1) (n_k+)$.

	X direction	Y direction	Z direction	# of values
Method #1:	*DI	*DJ	*ZCORN	$n_i + n_j + 8 n_i n_j n_k$
Method #2:	*XCORN	*YCORN	*ZCORN	$2 (n_i+1) (n_j+1) (n_k+1) + 8 n_i n_j n_k$
Method #3:	*COORD ¹	*COORD ¹	*ZCORN	$6 (n_i+1) (n_j+1) + 8 n_i n_j n_k$
Method #4:	*CORNERS ¹	*CORNERS ¹	*CORNERS ¹	$24 n_i n_j n_k$
Method #5:	*NODEX	*NODEY	*NODEZ	$3 n_{nodes}$
Method #6:	*NODES ¹	*NODES ¹	*NODES ¹	$3 n_{nodes}$
¹ – appears at most once per grid, e.g., one *NODES assigns corner points in all directions				

The following is a model for a corner point block, giving labels for its corners:

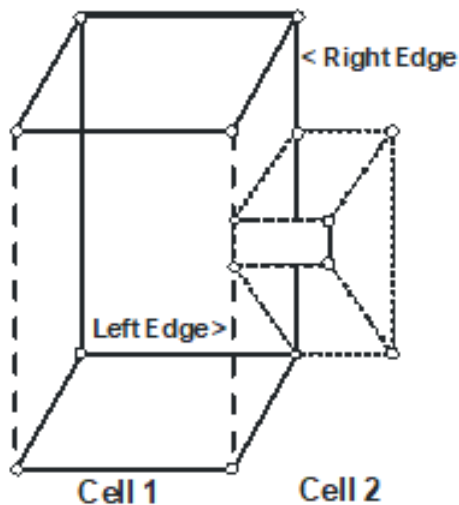


The block is the volume contained within the 6 faces, where each face is made by connecting the corner points with line segments as shown, and filling in the faces with a nonlinear (bilinear) interpolation. Faces need not be planar but should not be too far from planar. The numerical results will be better if the cell shape is kept close to a rectangle. The cell's pressure node (location) is at its barycentre (arithmetic average of its corners).

Even though the simulator performs tests to see if cells are not too distorted, examination of a grid with a visualization package is always recommended. Intrusions of one cell into another should be avoided.

Cells can make flow connections in two different ways. The usual situation is when four corners defining one face on one cell match (within a tolerance) the four corners defining a face on another cell. Projections (onto both sides) of the common face are used in the calculation of transmissibility for flow between the cells.

The other situation involves vertical faulting. If one of the side faces (not the top or bottom) of a cell is planar, and it overlaps a planar side face on another cell, and the two left and the two right edges of each face are collinear (that is, they lie on the same line), then a flow connection will be made based on the overlap area. An example follows with the shared edges marked:



A cell's vertical faces need not always be planar although this is often the case. Corner point grids defined using the ***COORD** keyword involve cells that share vertical edges, and if the ***COORD** lines are parallel, their side faces will be planar. The face overlap need occur only to within a tolerance.

The tolerance level for the above cases is controlled by ***CORNER-TOL**. If cells intrude on each other to too large a degree, then corner points and faces cannot make contact and flow will not occur.

EXAMPLES: